

Trial Test 8

- Submit your answers via Google Forms (you have only 1 hour and 10 mins to complete).
- Your score will be available immediately after the test. Please, check your emails.
- If you have technical issues, let me know in DM or in the Group Chat.
- You are not allowed to use the calculator. Also, do not google, be honest!
- Do not panic and manage your time wisely. Read the questions carefully and try to answer all of them.

Good Luck!

1) Evaluate

$$(\sqrt{27}-\sqrt{12}) * (\sqrt{49}+\sqrt{8})$$

a) $9\sqrt{3} - 2\sqrt{2}$

b) $5\sqrt{3} - \sqrt{6}$

c) $\sqrt{15} + \sqrt{57}$

d) $7\sqrt{3} + 2\sqrt{6}$

2) Find the sum of infinite geometric sequence if:

$$b_4 = 4, b_8 = 1/4$$

a) 64

b) 48

c) 32

d) 144

3) Find $3n+2m$

$$3^n * 243^{2n} = 3^{121}$$

$$3^m = \frac{1}{9}$$

a) 37

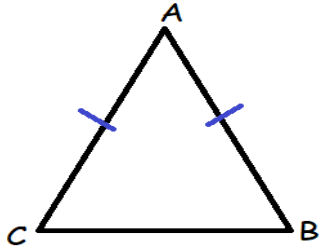
b) 29

c) 27

d) 39

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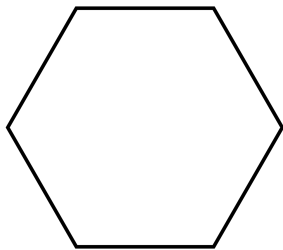
4) Find the area of the triangle, if given that $BC = 8$, $AB = AC = 6$ and $A = 120^\circ$



- a) $9\sqrt{3}$
- b) $12\sqrt{3}$
- c) $18\sqrt{3}$
- d) $6\sqrt{3}$

5) Given that a regular hexagon has a side length of 8.

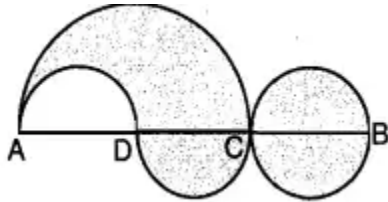
Find the area of a given polygon:



- a) $96\sqrt{3}$
- b) 64
- c) 96
- d) $64\sqrt{3}$

6) The given figure consists of three semi-circular arcs and one circle. The smaller arcs are equal. If $AB=9$ cm, $AC=6$ cm.

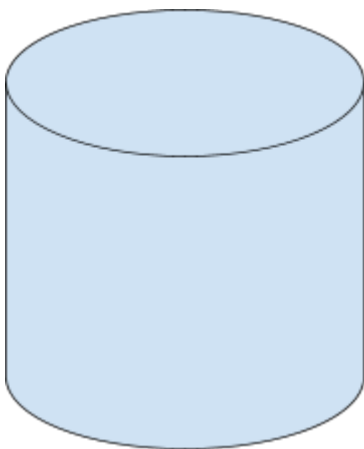
What is the complete length of the two figures?



- a) 9π
- b) 6π
- c) 3π
- d) 18π

7) Given that the height and radius of the cup are equal to 18 and 9, respectively.

Find the volume that is taken by water if it occupies $\frac{1}{2}$ of a cup.



- a) 1624π

b) 1458π

c) 729π

d) 2916π

8) Find the number of sides of a regular polygon, if each of its interior angle is 160°

a) 20

b) 18

c) 22

d) 16

9) Rearrange the formula in terms of R:

$$G = 12 + \sqrt{6(R + 2)^2 + 4}$$

a) $R = \sqrt{\frac{(G-12)^2 - 4}{6}} - 2$

b) $R = \sqrt{\frac{(G-12)^2 + 4}{6}} - 2$

c) $R = \sqrt{\frac{(G-12)^2 - 4}{6}} + 2$

d) $R = \sqrt{\frac{(G-12)^2 + 4}{6}} + 2$

10) How many times do the function $f(x) = x^4 + 5x^3 - x^2 - 5x$ intercept the x-axis?

a) 1

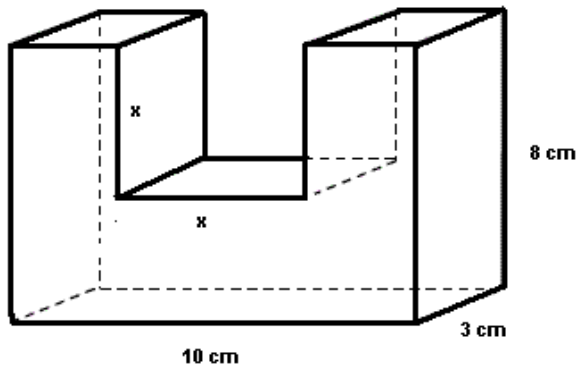
b) 2

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c) 3

d) 4

11) Find the value of x , if the volume of the given figure is equal to 165 cm^3



a) 4

b) 7

c) 5

d) 6

12) Convert the sum from decimal to fractional form:

$$0.(2) + 0.(63)$$

a) $47/99$

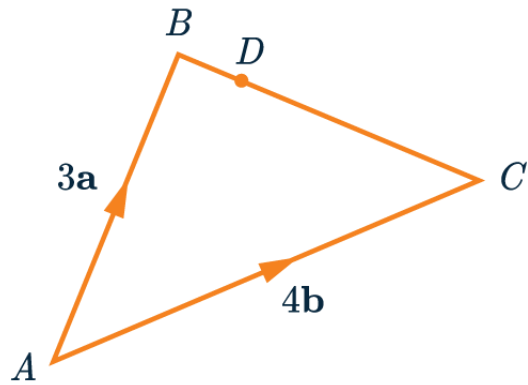
b) $8/33$

c) $85/99$

d) $19/66$

13) Write the vector $\vec{AD}$ in terms of $\vec{a}$ and $\vec{b}$, if the ratio $BD:DC = 1:3$

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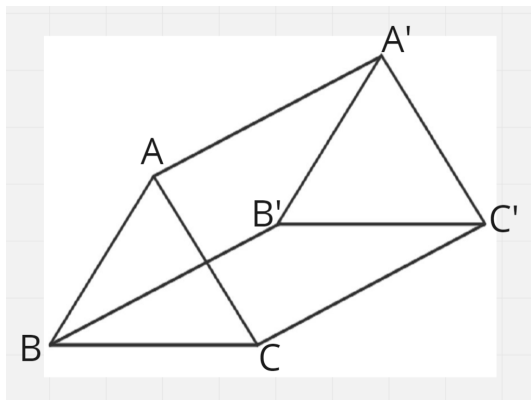
a) $\frac{9a}{2} + b$

b) $\frac{9a}{2} - b$

c) $\frac{9a}{4} + b$

d) $\frac{9a}{4} - b$

14) Find the surface area of regular triangular prism, if given that $BC=AC=AB=6$ and $AA'=10$



a) $180 + 18\sqrt{3}$

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b) $120 + 18\sqrt{3}$

c) $180 + 16\sqrt{3}$

d) $120 + 16\sqrt{3}$

15) Convert 17280 km/h into m/s:

a) 4800

b) 2880

c) 2400

d) 1440

16) A fisherman leaves his home port and heads in the direction $N70^\circ W$. He travels 30 mi and reaches Egg Island. The next day he sails $N10^\circ E$ for 50 mi, reaching Forrest Island. ($\sin(100) = 0.985$, $\cos(100) = -0.17$)

Find the bearing from Forrest Island back to his home

port

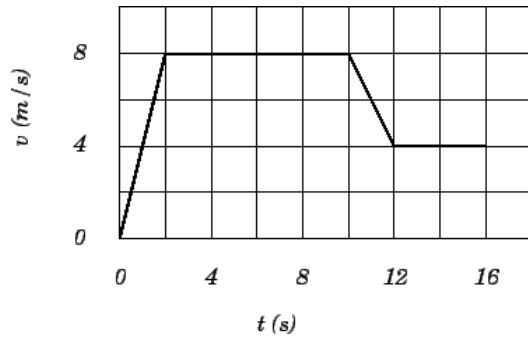
a) 18.2

b) 20.6

c) 22.4

d) 24.8

17) Find the total distance using the following graph



- a) 100
- b) 110
- c) 120
- d) 130

18) Solve the inequality

$$\frac{x^4 - 3x^3 - 6x^2 + 8x}{(x-1)^2} > 0$$

- a) $(-\infty, -2) \cup (0, 1) \cup (4, \infty)$
- b) $(-\infty, -1) \cup (0, 1) \cup (6, \infty)$
- c) $(-\infty, -2) \cup (0, 2) \cup (6, \infty)$
- d) $(-\infty, -1) \cup (0, 2) \cup (4, \infty)$

19) A tank is $\frac{3}{4}$ filled with water. When 300 ml of water is poured out of the tank, the tank becomes $\frac{3}{5}$ full.

What is the capacity of the tank in liters?

- a) 1
- b) 1.5

c) 2

d) 2.5

20) How many liters of 5% salt mixture should be added to 100ml of 20% salt mixture to have 10% mixture

a) 0.1

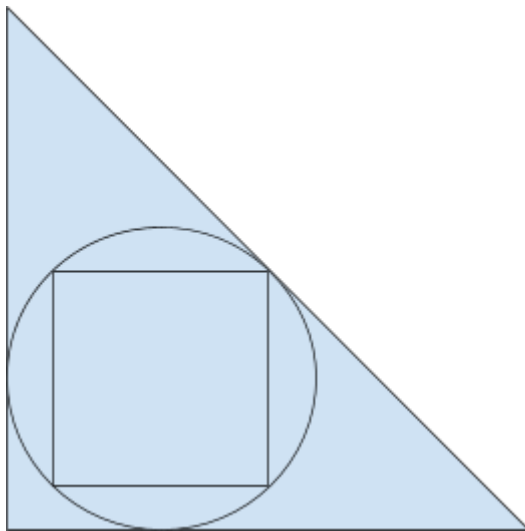
b) 0.15

c) 0.2

d) 0.25

21) Given that the radius of the circle is equal to 5, and the legs of the triangle are equal to 20.

What is the ratio of area of square to area of triangle



(figure is not to scale)

a) $\frac{1}{4}$

b) $\frac{1}{6}$

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c) $\frac{1}{8}$

d) $\frac{1}{3}$

22) $a \bullet b = a^2 + b^2$, $a \diamond b = a + 2b$.

Find the value of $(2 \bullet 4) \bullet (3 \diamond 1)$

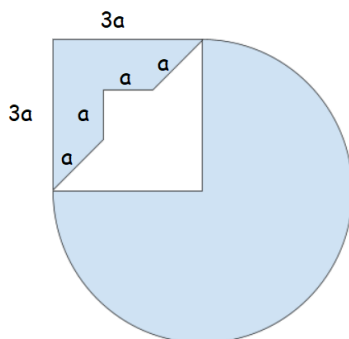
a) 545

b) 425

c) 385

d) 295

23) Find the total shaded area in terms of a



a) $a^2 \left(\frac{27\pi}{4} + 4 \right)$

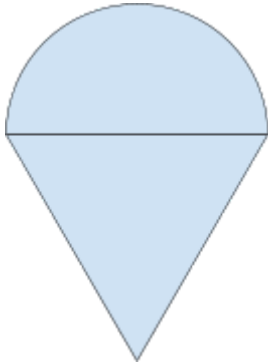
b) $a^2 \left(\frac{27\pi}{4} + 2 \right)$

c) $a^2 \left(\frac{27\pi}{2} + 2 \right)$

d) $a^2 \left(\frac{27}{4} + 2 \right)$

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24) Find the area of ice cream, if given that h is equal to 15 while radius of circle is equal to 5. (hint: height of triangle is equal to total height subtracted by radius)



a) $50 + \frac{25\pi}{2}$

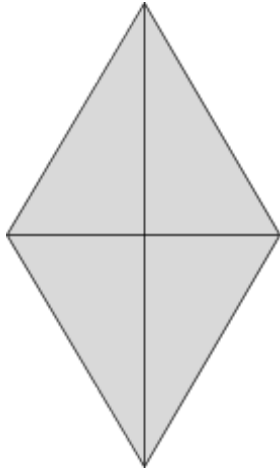
b) $50 + \frac{27\pi}{2}$

c) $25 + \frac{25\pi}{2}$

d) $25 + \frac{27\pi}{2}$

25) Given that $d_1 = x$, $d_2 = x - 2$, and area of rhombus is equal to 24

Find the value of the sides of rhombus



- a) 4
- b) 5
- c) 6
- d) 7

26) Find the roots of equation $x^4 - 2x^3 - 45x^2 - 34x + 80$, if given that one of the roots is 1

- a) $-5, -2, 1, 8$
- b) $-4, 1, 5, 8$
- c) $-8, -5, 1, 4$
- d) $-8, 2, 1, 5,$

27) Find an equation of the line perpendicular to the line $4x - 6y + 7 = 0$ passing through (3,4).

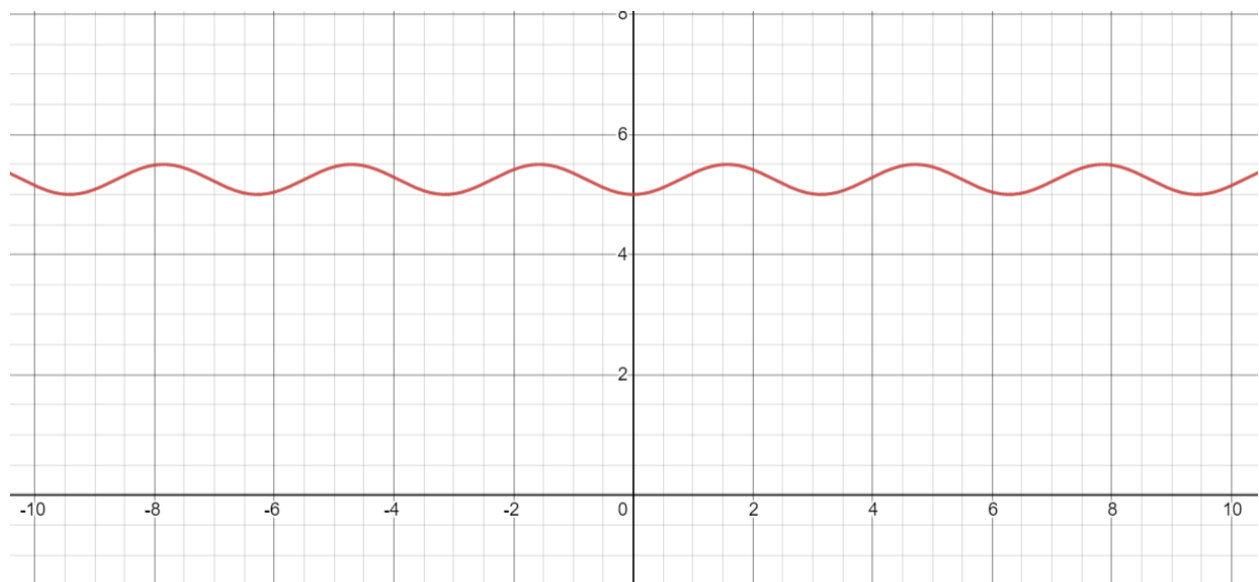
- a) $y = -\frac{3}{2}x + \frac{17}{2}$

$$\text{b) } y = -\frac{3}{2}x - \frac{17}{2}$$

$$\text{c) } y = \frac{3}{2}x + \frac{17}{2}$$

$$\text{d) } y = \frac{3}{2}x - \frac{17}{2}$$

28) Recognise the given function:



$$\text{a) } y = \sin^2(x) + 5$$

$$\text{b) } y = \frac{1}{2}\sin^2(x) + 5$$

$$\text{c) } y = \cos^2(x) + 5$$

$$\text{d) } y = \frac{1}{2}\cos^2(x) + 5$$

29) Give your answer as an ordinary number, correct to 3 significant figures.

$$\frac{4.256 \cdot 10^6 - 4.256 \cdot 10^4}{2.128 \cdot 10^{11}}$$

a) 0.0000198

b) 0.0000298

c) 0.0000398

d) 0.0000498

30) Factor 738 using prime numbers:

a) $2^1 * 3^3 * 41^1$

b) $2^1 * 3^2 * 41^1$

c) $2^2 * 3^2 * 41^1$

d) $2^1 * 3^2 * 83^1$